

Linear algebra review

COMPSCI 753

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Slides adapted from Kaiqi Zhao

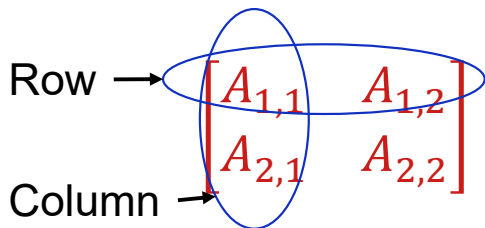
Basic Operands

Scalar

- A scalar is a single number
- Integers, real numbers, etc.
- Denoted by a single letter like c or N

Matrices

- A matrix is a 2-D array of numbers:
- Denoted by bold capital $A \in \mathbb{R}^{m \times n}$



Vectors

- A vector is a 1-D array of numbers:
- Can be real, binary, integer, etc.
- Denote by bold font like $v \in \mathbb{R}^n$

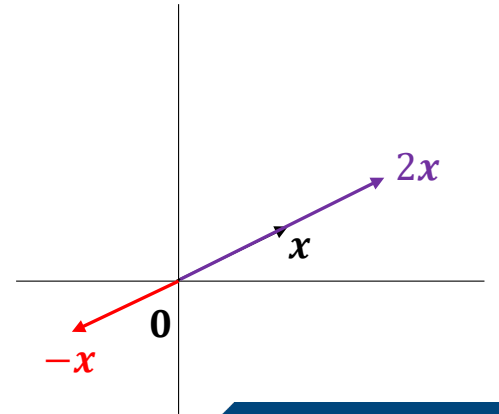
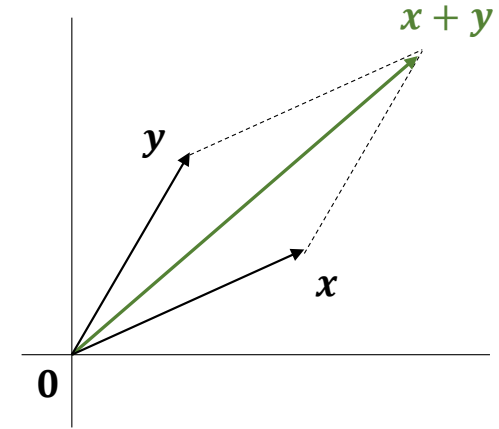
$$v = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

Tensors

- A tensor is an array of numbers, that may have:
 - zero dimensions, and be a scalar
 - one dimension, and be a vector
 - two dimensions, and be a matrix
 - or more dimensions.

Vector Operation – Norms

- Functions that measure how “large” a vector is
- A norm satisfies the following properties:
 - $f(\mathbf{x}) = 0 \Rightarrow \mathbf{x} = \mathbf{0}$ (positive definite)
 - $f(\mathbf{x} + \mathbf{y}) \leq f(\mathbf{x}) + f(\mathbf{y})$ (triangle inequality)
 - $\forall a \in \mathbb{R}, f(a\mathbf{x}) = |a|f(\mathbf{x})$ (absolutely scalable)

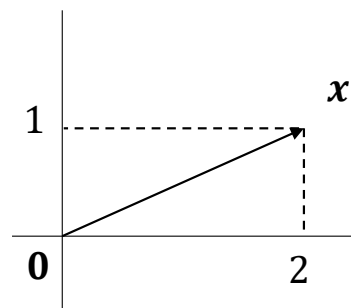


L_p Norm

$$\|x\|_p = \left(\sum_i |x_i|^p \right)^{\frac{1}{p}}$$

- L_1 norm, $p = 1$: $\|x\|_1 = \sum_i |x_i|$
- L_2 norm, $p = 2$: $\|x\|_2 = \sqrt{\sum_i x_i^2}$
- Max norm, infinite p : $\|x\|_\infty = \max_i |x_i|$

Example:



$$\|x\|_1 = ?$$

$$\|x\|_2 = ?$$

$$\|x\|_\infty = ?$$

By default, we use $\|\cdot\|$ or $|\cdot|$ to denote L_2 norm $\|\cdot\|_2$ in the lectures

Vector Operation – Dot Product

- Dot product of two vectors

$$\mathbf{a} \cdot \mathbf{b} = \sum_i a_i b_i$$

- It can also be written in terms of their L_2 norms and the angle θ between them

$$\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos(\theta)$$

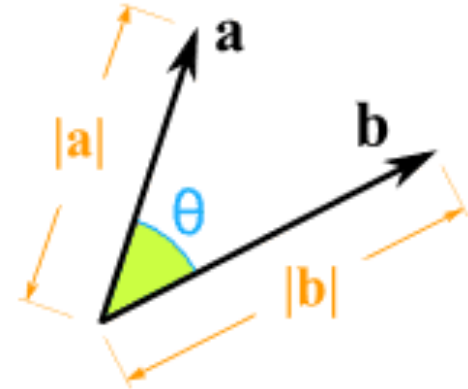


Image taken from math is fun

Vector projection

- Given two vectors $\mathbf{a}$ and $\mathbf{b}$, let $\hat{\mathbf{b}} = \frac{\mathbf{b}}{\|\mathbf{b}\|}$ be the unit vector in the direction of $\mathbf{b}$.
- Then $\mathbf{a} \cdot \hat{\mathbf{b}}$ gives a projection of $\mathbf{a}$ onto the line of $\mathbf{b}$, i.e.,

$$\mathbf{a} \cdot \hat{\mathbf{b}} = \mathbf{a} \cdot \frac{\mathbf{b}}{\|\mathbf{b}\|} = \|\mathbf{a}\| \cos(\theta)$$

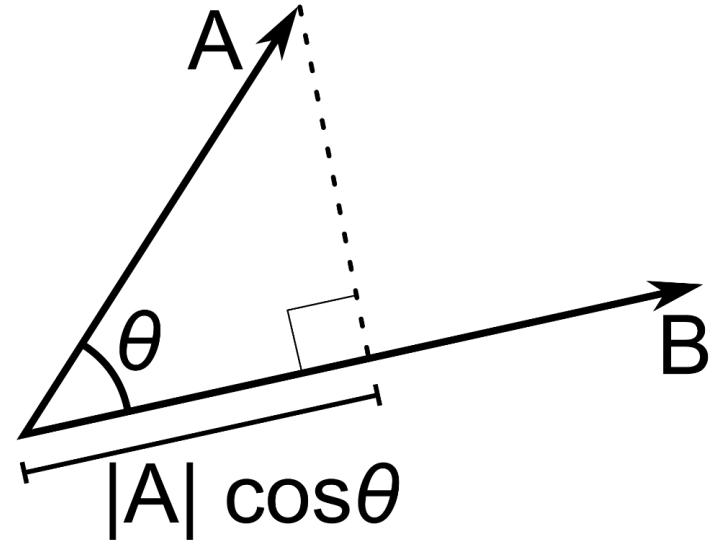


Image taken from Wikipedia

Cosine Similarity

- Cosine of the angle between two vectors measures similarity:

$$\cos(\theta) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}$$

- **Orthogonal vectors $\mathbf{a} \perp \mathbf{b}$** : Two vectors $\mathbf{a}$ and $\mathbf{b}$ are orthogonal to each other if $\mathbf{a} \cdot \mathbf{b} = 0$

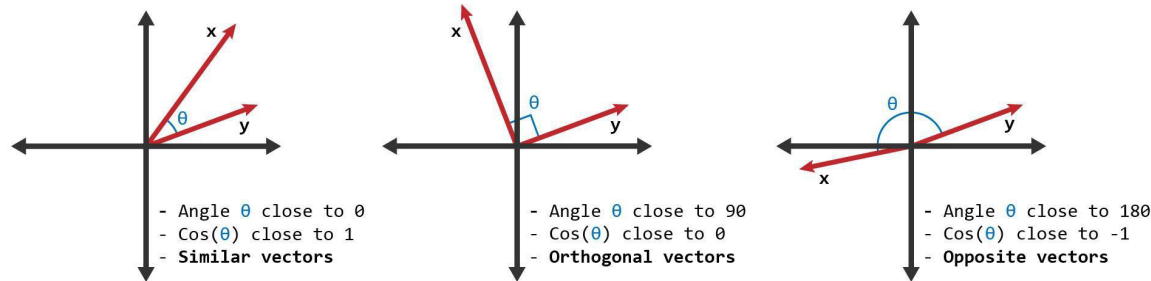


Image taken from:

<https://www.learndatasci.com/glossary/cosine-similarity/>

Matrix Operation – Transpose 转置

$$(A^T)_{i,j} = A_{j,i}$$

$$A = \begin{bmatrix} A_{1,1} & A_{1,2} \\ A_{2,1} & A_{2,2} \\ A_{3,1} & A_{3,2} \end{bmatrix} \Rightarrow A^T = \begin{bmatrix} A_{1,1} & A_{2,1} & A_{3,1} \\ A_{1,2} & A_{2,2} & A_{3,2} \end{bmatrix}$$

Example:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}^T = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix}$$

- The transpose of the matrix can be thought of as a mirror image across the main diagonal.

Example:

$$(AB)^T = B^T A^T$$
$$\left(\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right)^T = \begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix}$$

Matrix Operation – Dot-Product

$$\mathbf{C} = \mathbf{A} \cdot \mathbf{B}$$

$$C_{i,j} = \sum_k A_{i,k} B_{k,j}$$

$$\begin{bmatrix} A_{1,1} & A_{1,2} & A_{1,3} \\ A_{2,1} & A_{2,2} & A_{2,3} \end{bmatrix} \begin{bmatrix} B_{1,1} & B_{1,2} & B_{1,3} \\ B_{2,1} & B_{2,2} & B_{2,3} \\ B_{3,1} & B_{3,2} & B_{3,3} \end{bmatrix} = \begin{bmatrix} C_{1,1} & C_{1,2} & C_{1,3} \\ C_{2,1} & C_{2,2} & C_{2,3} \end{bmatrix}$$

- The **#columns** of A has to be the same as **#rows** in B.
- Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{B} \in \mathbb{R}^{n \times p}$ then $\mathbf{C} \in \mathbb{R}^{m \times p}$

Question: How about a matrix multiplies a vector, e.g., $\mathbf{A}\mathbf{x}$?

Matrix Operation – Dot-Product with Vector

$$c = A \cdot v \qquad c_i = \sum_k A_{i,k} v_k$$

$$\begin{bmatrix} A_{1,1} & A_{1,2} & A_{1,3} \\ A_{2,1} & A_{2,2} & A_{2,3} \end{bmatrix}_{m \times n} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}_{n \times 1} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}_{m \times 1}$$

- The **#columns** of A has to be the same as **#rows** in v.

Matrix Operation – Inversion

- Matrix inverse: $A^{-1} A = I$ $I = \begin{bmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{bmatrix}$
- Solving a system using an inverse:

$$Ax = b \Rightarrow A^{-1}Ax = A^{-1}b \Rightarrow x = A^{-1}b$$

- Matrix can only be inverted if it is a **square matrix** and all the rows/columns are linearly independent

- Orthogonal matrix:**

$$A^{\top} A = A A^{\top} = I$$
$$A^{-1} = A^{\top}$$

Are the following matrices orthogonal?

$$\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Matrix Operation – Determinant

- Determinant of a square matrix maps the matrix to a scalar, denoted by $\det(\mathbf{A})$ or $|\mathbf{A}|$

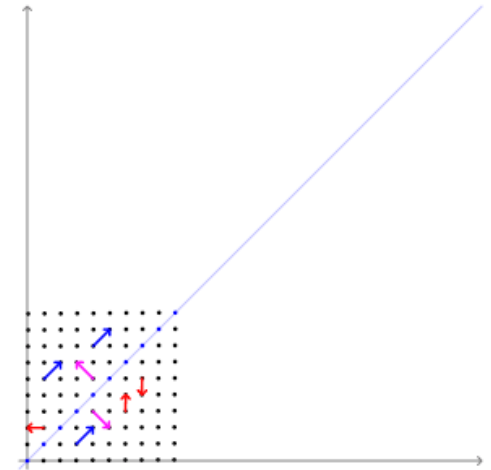
$$|\mathbf{A}| = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$

Image taken from Wikipedia

- Measures how much multiplication by the matrix expands or contracts the space.
- Useful Properties: $\det(\mathbf{AB}) = \det(\mathbf{A}) \det(\mathbf{B})$
 $\det(\mathbf{A}^\top) = \det(\mathbf{A})$

Matrix can be considered as a linear transformation of a space

$$\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$



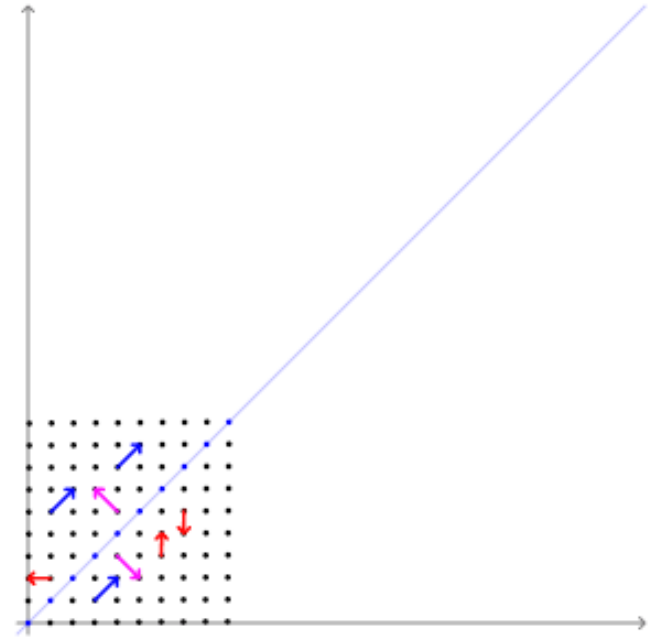
Eigenvalues and Eigenvectors

- An **eigenvector** of a square matrix A is a **nonzero** vector v such that left multiply by A only changes the scale of v .

$$Av = \lambda v$$

- The scalar λ is known as the **eigenvalue**.
- If we rescale the eigenvector v by a non-zero constant c , is cv an eigenvector of A ?
- We often use eigenvectors of unit length:

$$\|v\|_1 = 1 \quad \text{or} \quad \|v\|_2 = 1$$



Characteristic Polynomial

- Eigenvalue equation of matrix A :

$$A\mathbf{v} = \lambda\mathbf{v} \quad \Rightarrow \quad A\mathbf{v} - \lambda\mathbf{v} = 0 \quad \Rightarrow \quad (A - \lambda I)\mathbf{v} = 0$$

- If **nonzero** eigenvector $\mathbf{v}$ exists:

$$\det(A - \lambda I) = 0$$

- By expanding the determinant as a function of λ , we often get:

$$\det(A - \lambda I) = (\lambda_1 - \lambda)(\lambda_2 - \lambda) \dots (\lambda_n - \lambda) = 0$$

- The $\lambda_1, \lambda_2, \dots, \lambda_n$ are roots of the characteristic polynomial, and are eigenvalues of A

Example

- Consider the matrix: $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$
- The characteristic polynomial is:

$$\det(A - \lambda I) = \det \begin{bmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{bmatrix} = 3 - 4\lambda + \lambda^2 = 0$$

- It has roots $\lambda = 1$ and $\lambda = 3$ which are the two eigenvalues of A .
- We can then solve for eigenvectors using $A\mathbf{v} = \lambda\mathbf{v}$:

$$\mathbf{v}_{\lambda=1} = [1, -1]^\top \quad \text{and} \quad \mathbf{v}_{\lambda=3} = [1, 1]^\top$$

Calculate determinant of larger matrices:

<https://www.mathsisfun.com/algebra/matrix-determinant.html>

Eigen decomposition

- Suppose that $n \times n$ matrix A has n linearly independent eigenvectors $\{\mathbf{v}^{(1)}, \dots, \mathbf{v}^{(n)}\}$ with eigenvalues $\{\lambda_1, \dots, \lambda_n\}$.
- Concatenate eigenvectors to form matrix V .
- Concatenate eigenvalues to form vector $\boldsymbol{\lambda} = [\lambda_1, \dots, \lambda_n]^\top$
- The eigen decomposition of A is given by:

$$A = V \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} V^{-1} = V \text{diag}(\boldsymbol{\lambda}) V^{-1}$$